

RALEIGH COUNTY AP, WV, USA

WMO: 724120

Lat: 37.784N

Lon: 81.123W

Elev: 2514

StdP: 13.41

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03872

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.4	11.1	-3.9	4.9	12.0	0.7	6.2	15.7	23.0	36.5	20.4	34.7	7.1	310	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	84.8	69.4	82.8	68.7	80.9	68.0	72.5	80.3	71.3	78.6	70.1	77.5	7.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	121.8	76.2	68.9	116.5	75.0	67.8	112.0	73.9	37.8	80.6	36.6	78.8	35.6	77.2	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.1	17.0	15.4	DB	-0.9	88.0	6.8	2.9	-5.8	90.1	-9.8	91.8	-13.7	93.4	-18.6	95.5	
			WB	-1.7	74.9	6.8	1.8	-6.6	76.2	-10.6	77.3	-14.4	78.3	-19.4	79.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.5	31.5	34.8	42.2	53.3	60.9	67.9	70.7	69.9	64.2	54.0	43.6	35.8
	DBStd	16.46	12.50	11.38	11.21	9.70	7.74	5.03	4.34	4.31	6.49	8.72	9.87	10.75
	HDD50	2161	584	438	293	77	12	0	0	0	2	62	240	453
	HDD65	5160	1037	845	707	361	171	31	9	10	90	352	643	904
	CDD50	3073	12	13	52	177	349	536	642	618	429	184	47	14
	CDD65	597	0	0	1	11	43	117	186	163	67	9	0	0
	CDH74	3452	0	0	7	89	275	646	1065	904	414	50	2	0
	CDH80	592	0	0	0	6	30	95	209	163	82	7	0	0
Wind	WSAvg	6.8	8.4	8.3	8.2	8.1	6.6	5.6	5.0	4.8	5.4	6.3	7.1	7.6
Precipitation	PrecAvg	42.1	3.1	2.8	3.7	3.5	4.3	4.0	4.8	3.5	3.4	2.7	3.0	3.2
	PrecMax	57.2	6.4	7.0	9.2	7.6	7.6	8.1	9.6	8.0	10.0	5.9	6.7	6.4
	PrecMin	32.6	0.6	0.7	1.5	0.3	1.0	1.3	1.7	1.1	0.5	0.2	0.7	0.6
	PrecStd	5.9	1.4	1.3	1.7	1.4	1.9	1.7	1.9	1.4	2.0	1.4	1.4	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	66.6	74.3	80.6	83.3	85.5	87.6	86.4	86.3	78.8	71.1	64.5
		MCWB	54.3	54.9	57.5	59.2	65.3	69.1	71.7	70.2	66.4	64.5	57.7	55.8
	2%	DB	59.0	60.5	68.6	76.7	79.9	82.6	84.5	83.9	82.3	74.3	65.8	59.8
		MCWB	53.2	51.2	54.3	58.3	64.2	68.8	70.8	69.9	66.4	61.6	54.3	53.5
	5%	DB	54.6	56.3	63.8	73.1	77.1	80.4	82.3	81.8	79.0	71.0	62.3	56.1
		MCWB	49.7	49.0	51.9	56.5	63.1	68.0	69.7	69.2	66.2	59.4	53.3	50.9
	10%	DB	50.0	51.9	59.1	69.2	74.1	78.0	80.1	79.3	75.4	67.6	58.7	52.4
		MCWB	44.7	45.5	49.5	55.2	62.0	66.9	69.0	68.4	64.3	58.6	51.5	47.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.1	57.8	59.1	63.4	69.1	72.8	74.7	73.8	70.5	68.0	60.8	58.1
		MCDB	61.2	63.2	70.9	74.7	79.1	80.8	82.9	81.7	79.6	74.2	67.5	62.1
	2%	WB	54.0	54.0	56.5	61.2	67.1	70.8	72.9	72.3	69.0	65.0	58.0	55.0
		MCDB	58.2	58.5	65.1	71.5	75.8	78.9	80.8	79.8	77.3	71.4	63.0	58.6
	5%	WB	50.3	50.1	54.3	59.3	65.4	69.4	71.5	70.9	67.8	62.4	55.4	52.0
		MCDB	54.1	54.6	61.2	68.4	73.5	77.5	78.8	78.1	74.9	67.9	60.0	55.5
	10%	WB	45.7	45.9	51.5	57.2	63.7	68.3	70.5	69.6	66.3	60.0	52.5	48.0
		MCDB	49.1	50.8	57.2	65.6	71.0	75.6	77.4	76.7	72.7	65.5	57.3	51.6
Mean Daily Temperature Range	5% DB	MDBR	15.2	16.2	18.0	20.2	18.5	17.3	16.6	16.9	17.8	18.3	17.1	14.6
		MCDBR	18.2	22.1	23.2	24.1	21.8	19.9	19.2	19.5	21.8	21.8	20.8	18.4
	5% WB	MCWBR	13.0	15.4	13.1	11.6	9.8	9.0	8.0	8.4	9.2	10.9	12.7	13.7
		MCWBR	16.4	19.4	19.2	19.8	18.1	17.2	16.2	16.3	17.2	17.1	17.3	17.2
Clear-Sky Solar Irradiance	taub	taub	0.290	0.301	0.324	0.362	0.405	0.423	0.455	0.460	0.361	0.335	0.304	0.285
		taud	2.516	2.474	2.467	2.378	2.248	2.229	2.169	2.128	2.409	2.482	2.547	2.557
	Ebn at Noon	Ebn at Noon	287	297	299	291	279	272	263	258	282	280	279	282
		Edh at Noon	24	29	32	37	43	44	47	48	34	29	24	22
All-Sky Solar Radiation	RadAvg	RadAvg	618	834	1184	1549	1730	1904	1787	1664	1401	1050	765	545
	RadStd	RadStd	64	112	88	122	141	124	105	84	137	131	88	59

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	-265	+270	+119
(47)	Regional (30 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+247	+128

Nomenclature: See separate page